Revisiting Music Encoding for Music-to-Text Large Language Models: *What Is Encoded and What Is "Heard"*

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TAKE-HOME MESSAGE

MERT embeddings robustly **encode** single-note attributes, but LLMs **retrieve** only a limited portion of this information when answering in natural language.

MOTIVATION

Do LLMs actually access the musical information encoded in audio embeddings?

DATASET

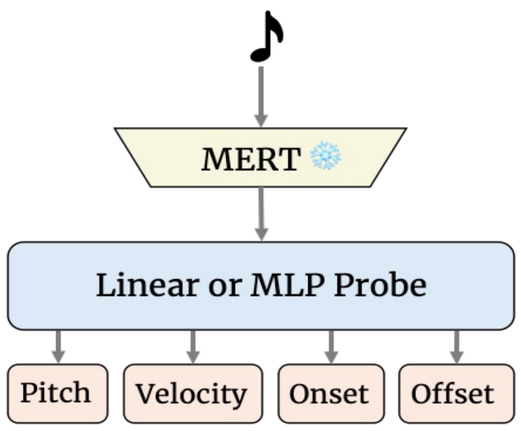
4,224 isolated piano notes, from MAPS ISOL subset (Emiya et al., 2010)

Pitch / Velocity / Onset / Offset

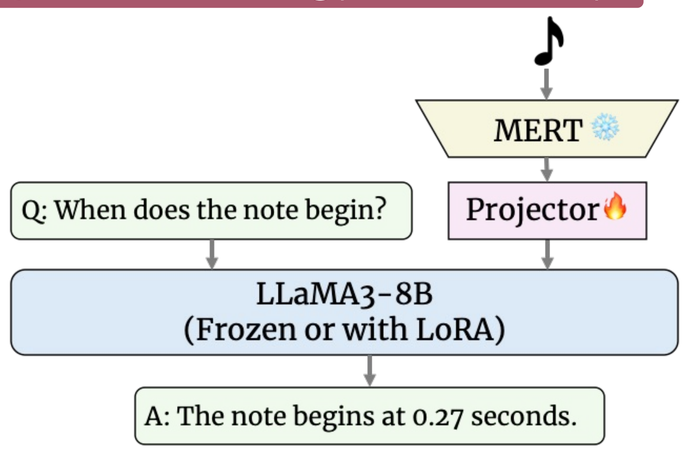
METHOD OVERVIEW

We examine the same embeddings from two perspectives.

A. Probing *(What is Encoded?)*

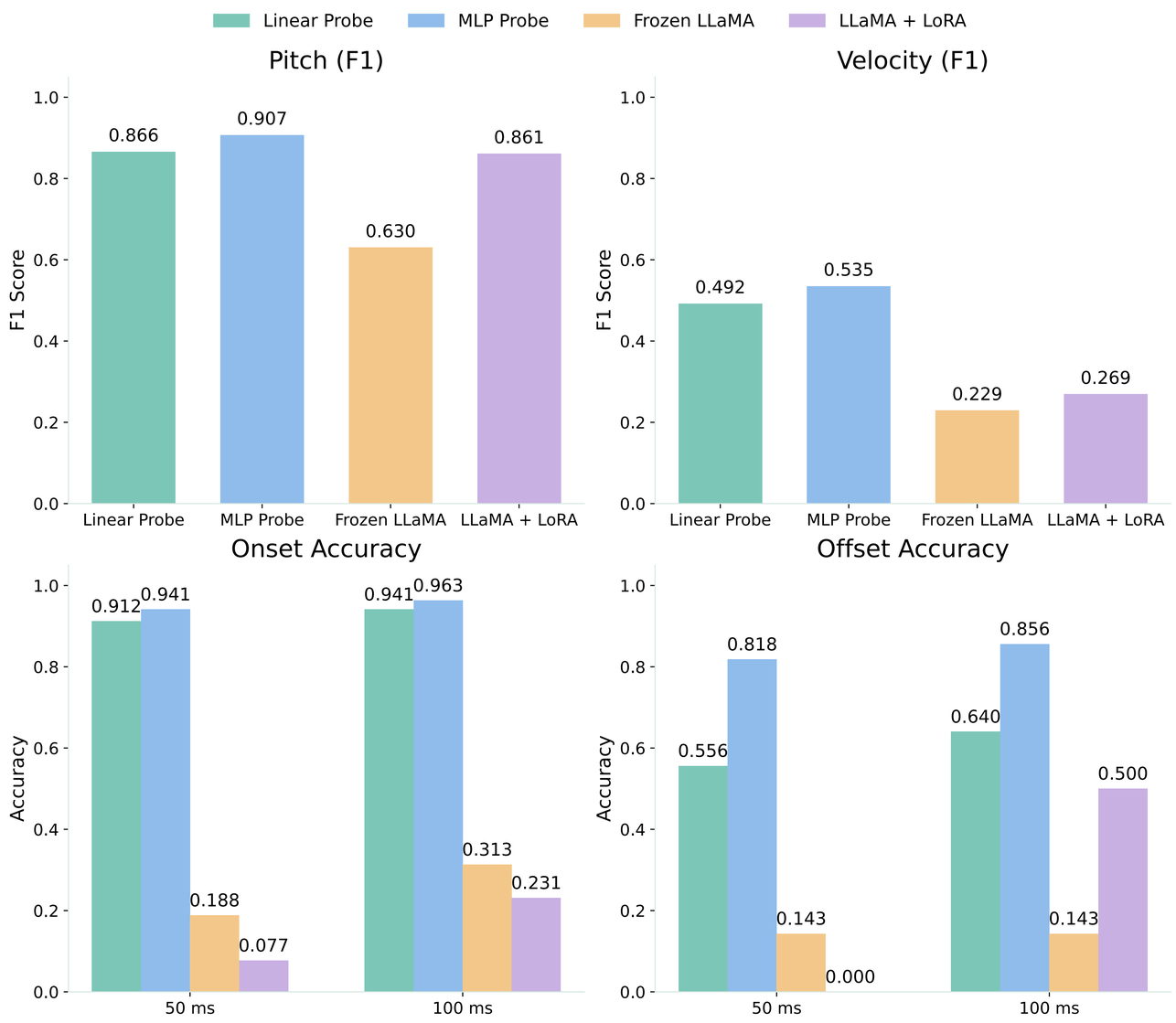


B. LLM Question-Answering *(What is Heard?)*



RESULTS

Performance Across Four Note-Level Attributes



KEY FINDINGS

- Probing decodes more information across all attributes.
- Onset and offset information degrade the most.
- Pitch remains the most accessible attribute for LLM-based answering.
- Fine-tuning improves stability but does not recover the missing information.

CONCLUSIONS

Encoded ≠ *Heard*

- Neural audio embeddings encode single-note attributes with high decodability.
- LLMs can access only a portion of this information in natural-language answering.
- Reliable handling of fundamental music tasks should be an explicit prerequisite before trusting their performance on more complex music-to-text tasks.

ACKNOWLEDGEMENTS

This work is supported in part by funding from the Social Sciences and Humanities Research Council of Canada (SSHRC) under Grant No. 895-2022-1004.

Social Sciences and Humanities Research Council of Canada
Conseil de recherches en sciences humaines du Canada